

2.4 Chapter Summary

The Main Ideas

- The exact numerical value of a quantity may not be known; the value is often given by an **approximation**.
- A numerical quantity is often given as the **limit** of a sequence of **successive approximations**.
- When a particular digit in a sequence of successive approximations **stabilizes**, that digit is assumed to appear in the limit.
- **Euler's method** is a procedure to construct a sequence of increasingly better **approximations** of a function defined by a set of rate equations and initial conditions. Each approximation is a piecewise linear function.
- The exact function defined by a set of rate equations and initial conditions can be expressed as the **limit** of a sequence of **successive Euler approximations** with smaller and smaller step sizes.

Expectations

- You should be able to use a program to **construct a sequence** of estimates for S , I , and R , given a specific S - I - R model with initial conditions.
- You should be able to **modify** the SIRVALUE and SIRPLOT programs to construct a sequence of estimates for the values of functions defined by other rate equations and initial conditions.
- You should be able to use programs that construct a sequence of **Euler approximations** for the function defined by a rate equation with an initial condition. The programs should provide both tabular and graphical output.
- You should be able to estimate the values of the roots of an equation $f(x) = 0$ using a **computer graphing package**.

- You should be able to find a square root using the **Babylonian algorithm**.
- You should be able to find the length of any piece of any curve.

Chapter Exercises

1. a) We have considered the logistic equation

$$y' = .1y \left(1 - \frac{y}{1000} \right)$$

On page 81 we looked at the resulting graph of y vs. t , using the starting value $y(0) = 100$. There is another graphical interpretation, though, that is also instructive. Note that the logistic equation specifies y' as a function of y . Sketch the graph of this function. That is, plot y values on the horizontal axis and y' values on the vertical axis; points on the graph will thus be of the form (y, y') , where the y' -coordinate is the value given by the logistic equation for the given y value. What shape does the graph have?

- b) For what value of y does this graph take on its largest value? Where does this y value appear in the graph of $y(t)$ versus t ?
- c) For what values of y does the graph of y' versus y cross the y -axis? Where do these y values appear in the graph of $y(t)$ versus t ?

Grids on Graphs

When writing a graphing program it is often useful to have the computer draw a grid on the screen. This makes it easier to estimate numerical values, for instance. We can use a simple FOR–NEXT loop inside a program to do this. For instance, suppose we had written a program (SIRPLOT or SEQUENCE, for instance) with the graphics already in place. Suppose the screen window covered values from 0 to 100 horizontally, and 0 to 50,000 vertically. If we wanted to draw 21 vertical lines (including both ends) spaced 5 units apart and 11 horizontal lines spaced 5,000 units apart, the following two loops inserted in the program would work:

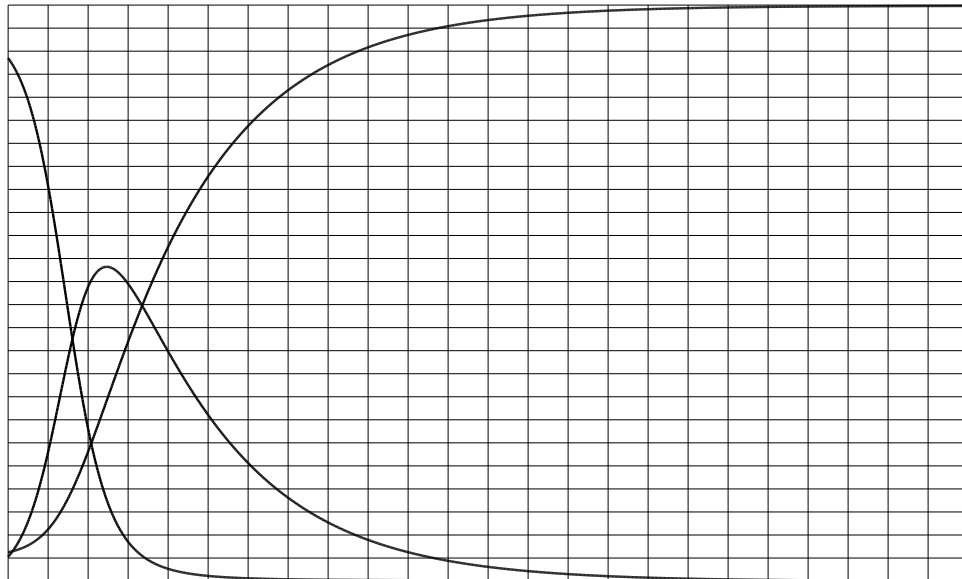
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FOR k = 0 TO 20
    Plot the line from (5 * k, 0) to (5 * k, 50000)
NEXT k
FOR k = 0 TO 10
    Plot the line from (0, 5000 * k) to (100, 5000 * k)
NEXT k

```

You should make sure you see how these loops work and that you can modify them as needed.

2. If the screen window runs from -20 to 120 horizontally, and 250 to 750 vertically, how would you modify the loops above to create a vertical grid spaced 10 units apart and a horizontal grid spaced 25 units apart?
3. Go back to our basic S - I - R model. Modify SIRPLOT to calculate and plot on the same graph the values of $S(t)$, $I(t)$, and $R(t)$ for t going from 0 to 120, using a stepsize of $\Delta t = .1$. Include a grid with a horizontal spacing of 5 days, and a vertical spacing of 2000 people. You should get something that looks like this:



Chapter 3

The Derivative

In developing the *S-I-R* model in chapter 1 we took the idea of the rate of change of a population as intuitively clear. The rate at which one quantity changes with respect to another is a central concept of calculus and leads to a broad range of insights. The chief purpose of this chapter is to develop a fuller understanding—both analytic and geometric—of the connection between a function and its rate of change. To do this we will introduce the concept of the **derivative** of a function.

3.1 Rates of Change

The Changing Time of Sunrise

The sun rises at different times, depending on the date and location. At 40° N latitude (New York, Beijing, and Madrid are about at this latitude) in the year 1990, for instance, the sun rose at

7:16 on January 23,
5:58 on March 24,
4:52 on July 25.

The time of sunrise
is a function

Clearly the time of sunrise is a function of the date. If we represent the time of sunrise by T (in hours and minutes) and the date by d (the day of the year), we can express this functional relation in the form $T = T(d)$. For example, from the table above we find $T(23) = 7:16$. It is not obvious from the table, but it is also true that the rate at which the time of sunrise is changing is different at different times of the year— T' varies as d varies.